MILITARY SPECIFICATION
FIN ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Fin Assembly which is used on the M74 Nike Hercules (Training).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWING

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments.
8846093 - Fin Assembly.
8846977 - Carton, Packing, Ammunition, for Fin, Atomic Warhead Section.
ET-8846093 - Equipment Tabulation.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The fin assembly shall comply with all requirements specified on the drawing (dwg) 8846093, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

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3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Weight. The weight of the fin assembly shall be a maximum of eight (8) pounds when measured in accordance with 4.4.1.

3.5 Finish. After the casting has been cured and has been removed from the mold, all flash and disfiguration marks shall be removed. Imperfections, which can not be removed without substantially changing over-all contour, shall be filled with the composition specified on dwg 8846093 and blended smooth.

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8846093) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

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Table I
First article inspection and tests

Inspection or Test	Dest	Non Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Weight (see 3.4)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Fin assembly (see dwg 8846093).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of threaded insert in base of fin (3 places)	Gage
102.	Pitch diameter of thread	Gage
103.	Minor diameter of thread, max	Gage
104.	Overall length	Gage
105.	Height from base to center of top surface of cylindrical projection	Gage
106.	Height from base to top surface of pitot tube	Gage
107.	Diameter of cylindrical projection	Gage
108.	Angle between the base and top edge of fin	Gage

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109. Surface finish (see 3.5).....Visual
110. Objective evidence relative to material missing or improper (see 3.1).....Visual
111. No cracks on any surface.....Visual

Minor:
201. Markings.....Visual
202. Workmanship (see 3.6).....Visual

4.3.2.1.2 Carton, packing, ammunition, for fin,atomic warhead section (see dwg 8846977 and section 5) before closing cover.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Item wrapped and secured with tape	Visual
	102. Item tight in carton	Manual
	103. Package material (see 3.1)	Visual
Minor:		
	201. Workmanship (see 3.6)	Visual/Manual

4.3.2.1.3 Carton, packing ammunition, for fin, atomic warhead section (see dwg 8846977 and section 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Carton sealed properly	Visual/Manual
	102. Label waterproofed	Visual
Minor:		
	201. Marking	Visual
	202. Workmanship (see 3.6)	Visual/Manual

4.3.2.1.4 Packing and marking for storage and shipments (see dwg 8808362 and section 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Contents tight in package	Manual

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102. Correctly packed.....Visual/Manual
103. Package material (see 3.1).....Visual

Minor:
201. Marking.....Visual
202. Workmanship (see 3.6).....Visual/Manual

4.3.3 Testing. All tests shall be performed.

Test	Test Classification
a. Weight	Major

This test is not considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Weight. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.1. If any item fails to comply with 3.4, the lot shall be rejected.

Table IV

Sample sizes for non-destructive testing

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
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24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). Els for assemblies are listed on the Equipment Tabulation, ET-8846093. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

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a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Weight. The weight of the fin assembly shall be measured utilizing an approved scale.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The fin assembly shall be packaged in accordance with dwg 8846977 using the specified moisture barrier.

5.1.2 Level C. The fin assembly shall be packaged in accordance with dwg 8846977 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The fin assembly shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwg 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.
a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

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6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

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Preparing activity:
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OMB Approval
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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